

Table II: Handheld DMMs from Different Manufacturers

	Fluke	Agilent	B&K Precision Corp	Motwane	Meco	Metrix+
Model	175	U1232A	2709B	M63	450 TRMS	18xL
Range	Voltage	600mV-1000V	600mV-600V	660mV-1000V	600mV/1000V	600mV-1000V
	Current	60mA-10A	60µA-10A	660µA-10A	60mA/20A	600µA-600A
	Resistance	600Ω-50MΩ	600Ω-60MΩ	660Ω-66MΩ	600Ω-60MΩ	600Ω-60MΩ
	Diode	2.4V	2V	3.5V		2.8V
	Capacitance	100nF-999µF	1000nF-10mF	6.6nF-66mF	40nF-400µF	40nF-400µF
	Frequency	99.99kHz	99.99kHz	66MHz	10MHz	10MHz
	Temperature (°C)	-40 to 400 degrees	-40 to 1372 degrees	NA	NA	-50 to 700 degrees
Resolution	6000 counts	6000 counts	6600 counts	6000 Counts	19999 counts	6000 counts
DC accuracy	0.15% + 2 counts	0.5% + 2 counts	0.5% rdg + 2 dgts	0.5% + 3 counts	0.05% rdg + 3 dgt	0.8% rdg + 10 dgt
Input impedance	10 M ohms	10 M ohms	10 M ohms	10 M ohms		
True RMS	Yes	Yes	Yes	Yes	Yes	Yes
Display updates	4 per second	5 per second	2 per second	3 per second		
Autorangeing	Yes	Yes	Yes	Yes	No	Yes
Capacitance meter	Yes	Yes	Yes	Yes	Yes	Yes
Frequency meter	Yes	Yes	Yes	Yes	Yes	Yes
Temperature measurement	Yes	Yes	No	No	No	Yes
Battery life	400 hr	500 hr	150 hr		150 hr	
Power consumption		450mVA maximum		4 mA		
CAT III	Yes	Yes	Yes			
Fuse	11 A, 1000V fast fuse	10 x 38 mm 11A/1000V fast-acting fuse		CAT IV/600 V		

DMMs with this function are provided with a K-type thermocouple cable to measure temperature. Check the DMM for maximum range of temperature in the datasheet and see if it suitable for your requirement.

Another range that you need to check is the diode function range. Choose the ones with maximum range. For example, if the diode function range is only 2V, you will not be able to test an LED with more than 2V forward voltage drop.

Resolution. The resolution of a DMM is the smallest reading that can be shown on the display. Resolution of the DMM used to be described in digits (such as 4.5, 5.5, etc). But these days, the resolution is specified in counts. For example, let us look at 4.5 digits, or 4 and 1/2 digits DMM. The '4' represents full digits, meaning digits that can take up values from 0 to 9 on the display. The '1/2' or '0.5' represents the first digit of the display and the maximum it can read. In this case, '1/2' digit can represent digits 0 or 1. Therefore 4.5 digits mean that the display can show 00000 to 19999, which means 20,000 counts. To calculate the smallest reading that can be shown on the display, just divide the full-scale reading by the number of counts. For a 6000-count DMM with

maximum DC voltage range as 600V, the resolution will be 600/6000 = 0.1V.

Accuracy. Though all the measurement features will have their corresponding accuracies mentioned in the datasheet, but first check basic DC voltage accuracy. Accuracy will be mentioned in percentage (such as 0.5 per cent). For regular electronics work, you do not need a DMM with very high accuracy (such as 0.05 per cent). You can do reasonably well with just 0.5 per cent accuracy. Accuracy in other measurement functions is also important, and it should be very close to basic DC voltage accuracy. Current accuracy is a different story as for current measurement a shunt resistor is used, which makes it difficult to achieve high accuracy.

Input impedance. Input impedance is an important specification. Buy the DMM that has input impedance as high as 10 mega-ohms. When you are measuring voltage across a component, you are effectively putting the input resistance of the DMM in parallel to the resistance of the component. Therefore small input impedance will change the values you are trying to measure.

True RMS. There are two types of DMMs available: 'Average-responding' and 'true RMS' (root mean square). True RMS is the measurement of AC voltage or current that reflects the amount of power dissipated by a resistive load driven by the equivalent DC value. An average responding AC DMM is calibrated to read the same as a true RMS meter for sine wave inputs only. For other waveform shapes, an average-responding meter will exhibit substantial errors.

Updating speed. Display update rate is an important specification. General-purpose DMMs are available with update rate of 5 per second. Higher the update rate, the better.

Battery life. A good battery life will make your life easier with electronics. Some advanced DMMs are really power hungry devices. For a good general-purpose DMM, the battery life should be 300 to 500 hours.

Once you have understood the specifications mentioned above and finalised the specifications you need, you can go ahead and buy the DMM of your choice. The selected multimeter will provide you the measurement confidence together with safety. ●

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